



## 1. Physicochemical Property

| Property         | Value   | Comment                                                                                                                                                                                           |
|------------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Molecular Weight | 477.09  | Contain hydrogen atoms. Optimal:100~600                                                                                                                                                           |
| Volume           | 430.298 | Van der Waals volume                                                                                                                                                                              |
| Density          | 1.109   | Density = MW / Volume                                                                                                                                                                             |
| nHA              | 13.0    | Number of hydrogen bond acceptors. Optimal:0~12                                                                                                                                                   |
| nHD              | 4.0     | Number of hydrogen bond donors. Optimal:0~7                                                                                                                                                       |
| nRot             | 6.0     | Number of rotatable bonds. Optimal:0~11                                                                                                                                                           |
| nRing            | 4.0     | Number of rings. Optimal:0~6                                                                                                                                                                      |
| MaxRing          | 14.0    | Number of atoms in the biggest ring. Optimal:0~18                                                                                                                                                 |
| nHet             | 13.0    | Number of heteroatoms. Optimal:1~15                                                                                                                                                               |
| fChar            | 0.0     | Formal charge. Optimal:-4 ~4                                                                                                                                                                      |
| nRig             | 25.0    | Number of rigid bonds. Optimal:0~30                                                                                                                                                               |
| Flexibility      | 0.24    | Flexibility = nRot /nRig                                                                                                                                                                          |
| Stereo Centers   | 5.0     | Stereo Centers. Optimal: ≤ 2                                                                                                                                                                      |
| TPSA             | 195.12  | Topological Polar Surface Area. Optimal:0~140                                                                                                                                                     |
| logS             | -3.256  | The logarithm of aqueous solubility value.                                                                                                                                                        |
| logP             | 0.972   | The logarithm of the n-octanol/water distribution coefficients at pH=7.4.                                                                                                                         |
| logD             | 1.378   | The logarithm of the n-octanol/water distribution coefficient.                                                                                                                                    |
| pka (Acid)       | 6.207   | Acid-base dissociation constant (pKa) value represents the strength of a drug molecule's acidity or basicity.                                                                                     |
| pka (Base)       | 4.517   | Acid-base dissociation constant (pKa) value represents the strength of a drug molecule's acidity or basicity.                                                                                     |
| Melting point    | 181.713 | The predicted melting point of a compound is expressed in degrees Celsius (°C).<br>Melting points below 25°C are classified as liquids, while melting points above 25°C are classified as solids. |
| Boiling point    | 307.73  | The predicted melting point of a compound is expressed in degrees Celsius (°C).<br>A normal boiling point below 25°C is categorized as a gas.                                                     |

## 2. Medicinal Chemistry

| Property | Value | Decision | Comment |
|----------|-------|----------|---------|
|----------|-------|----------|---------|

|                       |          |   |                                                                                                                                                                                                                                                                              |
|-----------------------|----------|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| QED                   | 0.198    | ● | <ul style="list-style-type: none"> <li>■ A measure of drug-likeness based on the concept of desirability;</li> <li>■ Attractive: &gt; 0.67;</li> <li>■ unattractive: 0.49~0.67;</li> <li>■ too complex: &lt; 0.34</li> </ul>                                                 |
| GASA                  | 1.0      | ● | <ul style="list-style-type: none"> <li>■ ES: Easy to synthesize; HS: Hard to synthesize;</li> <li>■ The output value represents the probability of being difficult to synthesize, ranging from 0 to 1.</li> </ul>                                                            |
| Synth                 | 4.0      | ● | <ul style="list-style-type: none"> <li>■ Synthetic accessibility score is designed to estimate ease of synthesis of drug-like molecules.</li> <li>■ SAScore <math>\geq 6</math>, difficult to synthesize; SAScore &lt;6, easy to synthesize</li> </ul>                       |
| Fsp3                  | 0.333    | ● | <ul style="list-style-type: none"> <li>■ The number of sp<sup>3</sup> hybridized carbons / total carbon count, correlating with melting point and solubility.</li> <li>■ Fsp<sup>3</sup> <math>\geq 0.42</math> is considered a suitable value.</li> </ul>                   |
| MCE-18                | 90.643   | ● | <ul style="list-style-type: none"> <li>■ MCE-18 stands for medicinal chemistry evolution.</li> <li>■ MCE-18 <math>\geq 45</math> is considered a suitable value.</li> </ul>                                                                                                  |
| NPscore               | 1.148    | - | <ul style="list-style-type: none"> <li>■ Natural product-likeness score.</li> <li>■ This score is typically in the range from -5 to 5.</li> <li>■ The higher the score is, the higher the probability is that the molecule is a NP.</li> </ul>                               |
| Lipinski Rule         | 0.0      | ● | <ul style="list-style-type: none"> <li>■ MW <math>\leq 500</math>; logP <math>\leq 5</math>; Hacc <math>\leq 10</math>; Hdon <math>\leq 5</math></li> <li>■ If two properties are out of range, a poor absorption or permeability is possible, one is acceptable.</li> </ul> |
| Pfizer Rule           | 0.0      | ● | <ul style="list-style-type: none"> <li>■ logP &gt; 3; TPSA &lt; 75</li> <li>■ Compounds with a high log P (&gt;3) and low TPSA (&lt;75) are likely to be toxic.</li> </ul>                                                                                                   |
| GSK Rule              | 1.0      | ● | <ul style="list-style-type: none"> <li>■ MW <math>\leq 400</math>; logP <math>\leq 4</math></li> <li>■ Compounds satisfying the GSK rule may have a more favorable ADMET profile</li> </ul>                                                                                  |
| Golden Triangle       | 0.0      | ● | <ul style="list-style-type: none"> <li>■ 200 <math>\leq</math> MW <math>\leq</math> 500; -2 <math>\leq</math> logD <math>\leq</math> 5</li> <li>■ Compounds satisfying the Golden Triangle rule may have a more favorable ADMET profile.</li> </ul>                          |
| PAINS                 | 0 alerts | - | frequent hitters, Alpha-screen artifacts and reactive compound 480 substructures (J Med Chem 201053:2719-40)                                                                                                                                                                 |
| ALARM NMR             | 3 alerts | - | Thiol reactive compounds.                                                                                                                                                                                                                                                    |
| BMS                   | 0 alerts | - | undesirable, reactive compounds 176 substructures (J Chem Inf Model 200646:1060-8)                                                                                                                                                                                           |
| Chelator Rule         | 1 alerts | - | Chelating compounds.                                                                                                                                                                                                                                                         |
| Colloidal aggregators | 0.972    | - | <ul style="list-style-type: none"> <li>■ Category 0: non-colloidal aggregators;</li> <li>■ Category 1: colloidal aggregators.</li> <li>■ The output value is the probability of being colloidal aggregators, within the range of 0 to 1.</li> </ul>                          |

|                       |       |   |                                                                                                                                                                                                                                            |
|-----------------------|-------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FLuc inhibitors       | 0.028 | ● | <ul style="list-style-type: none"> <li>Category 0: non-fLuc inhibitors;</li> <li>Category 1: fLuc inhibitors.</li> <li>The output value is the probability of being fLuc inhibitors, within the range of 0 to 1.</li> </ul>                |
| Blue fluorescence     | 0.066 | ● | <ul style="list-style-type: none"> <li>Category 0: non-blue fluorescence;</li> <li>Category 1: blue fluorescence.</li> <li>The output value is the probability of being blue fluorescence, within the range of 0 to 1.</li> </ul>          |
| Green fluorescence    | 0.548 | ● | <ul style="list-style-type: none"> <li>Category 0: non-green fluorescence;</li> <li>Category 1: green fluorescence.</li> <li>The output value is the probability of being green fluorescence, within the range of 0 to 1.</li> </ul>       |
| Reactive compounds    | 0.218 | ● | <ul style="list-style-type: none"> <li>Category 0: non-reactive compound;</li> <li>Category 1: reactive compound.</li> <li>The output value is the probability of being reactive compound, within the range of 0 to 1.</li> </ul>          |
| Promiscuous compounds | 0.201 | ● | <ul style="list-style-type: none"> <li>Category 0: non-promiscuous compound;</li> <li>Category 1: promiscuous compound.</li> <li>The output value is the probability of being promiscuous compound, within the range of 0 to 1.</li> </ul> |

### 3. Absorption

| Property            | Value  | Decision | Comment                                                                                                                                                                                                                                                                                                                           |
|---------------------|--------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Caco-2 Permeability | -6.01  | ●        | Optimal: higher than -5.15 Log unit                                                                                                                                                                                                                                                                                               |
| MDCK Permeability   | -5.289 | ●        | <ul style="list-style-type: none"> <li>low permeability: <math>&lt; 2 \times 10^{-6}</math> cm/s</li> <li>medium permeability: <math>2-20 \times 10^{-6}</math> cm/s</li> <li>high passive permeability: <math>&gt; 20 \times 10^{-6}</math> cm/s</li> </ul>                                                                      |
| PAMPA               | 0.999  | ●        | <ul style="list-style-type: none"> <li>The experimental data for Peff was logarithmically transformed (logPeff).</li> <li>Molecules with log Peff values below 2.0 were classified as low-permeability (Category 0), while those with log Peff values exceeding 2.5 were classified as high-permeability (Category 1).</li> </ul> |
| Pgp-inhibitor       | 0.0    | ●        | <ul style="list-style-type: none"> <li>Category 1: Inhibitor;</li> <li>Category 0: Non-inhibitor;</li> <li>The output value is the probability of being Pgp-inhibitor</li> </ul>                                                                                                                                                  |
| Pgp-substrate       | 0.03   | ●        | <ul style="list-style-type: none"> <li>Category 1: substrate;</li> <li>Category 0: Non-substrate;</li> <li>The output value is the probability of being Pgp-substrate</li> </ul>                                                                                                                                                  |
| HIA                 | 0.337  | ●        | <ul style="list-style-type: none"> <li>Human Intestinal Absorption</li> <li>Category 1: HIA+ (HIA <math>&lt; 30\%</math>);</li> <li>Category 0: HIA- (HIA <math>\geq 30\%</math>);</li> <li>The output value is the probability of being HIA+</li> </ul>                                                                          |

|            |       |   |                                                                                                                                                                                                                                                                                                                            |
|------------|-------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $F_{20\%}$ | 0.185 | ● | <ul style="list-style-type: none"> <li>■ 20% Bioavailability</li> <li>■ Category 1: <math>F_{20\%} +</math> (bioavailability &lt; 20%);</li> <li>■ Category 0: <math>F_{20\%} -</math> (bioavailability <math>\geq</math> 20%);</li> <li>■ The output value is the probability of being <math>F_{20\%} +</math></li> </ul> |
| $F_{30\%}$ | 0.887 | ● | <ul style="list-style-type: none"> <li>■ 30% Bioavailability</li> <li>■ Category 1: <math>F_{30\%} +</math> (bioavailability &lt; 30%);</li> <li>■ Category 0: <math>F_{30\%} -</math> (bioavailability <math>\geq</math> 30%);</li> <li>■ The output value is the probability of being <math>F_{30\%} +</math></li> </ul> |
| $F_{50\%}$ | 0.917 | ● | <ul style="list-style-type: none"> <li>■ 50% Bioavailability</li> <li>■ Category 1: <math>F_{50\%} +</math> (bioavailability &lt; 50%);</li> <li>■ Category 0: <math>F_{50\%} -</math> (bioavailability <math>\geq</math> 50%);</li> <li>■ The output value is the probability of being <math>F_{50\%} +</math></li> </ul> |

## 4. Distribution

| Property          | Value  | Decision | Comment                                                                                                                                                                                              |
|-------------------|--------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PPB               | 92.215 | ●        | <ul style="list-style-type: none"> <li>■ Plasma Protein Binding</li> <li>Optimal: &lt; 90%.</li> <li>■ Drugs with high protein-bound may have a low therapeutic index.</li> </ul>                    |
| VDss              | -0.107 | ●        | <ul style="list-style-type: none"> <li>■ Volume Distribution</li> <li>Optimal: 0.04-20L/kg</li> </ul>                                                                                                |
| BBB               | 0.016  | ●        | <ul style="list-style-type: none"> <li>■ Blood-Brain Barrier Penetration</li> <li>■ Category 1: BBB+; Category 0: BBB-;</li> <li>■ The output value is the probability of being BBB+</li> </ul>      |
| Fu                | 6.577  | ●        | <ul style="list-style-type: none"> <li>■ The fraction unbound in plasms</li> <li>■ Low: &lt;5%; Middle: 5~20%; High: &gt; 20%</li> </ul>                                                             |
| OATP1B1 inhibitor | 0.014  | ●        | <ul style="list-style-type: none"> <li>■ Category 0: Non-inhibitor; Category 1: inhibitor.</li> <li>■ The output value is the probability of being inhibitor, within the range of 0 to 1.</li> </ul> |
| OATP1B3 inhibitor | 0.966  | ●        | <ul style="list-style-type: none"> <li>■ Category 0: Non-inhibitor; Category 1: inhibitor.</li> <li>■ The output value is the probability of being inhibitor, within the range of 0 to 1.</li> </ul> |
| BCRP inhibitor    | 0.0    | ●        | <ul style="list-style-type: none"> <li>■ Category 0: Non-inhibitor; Category 1: inhibitor.</li> <li>■ The output value is the probability of being inhibitor, within the range of 0 to 1.</li> </ul> |
| MRP1 inhibitor    | 0.999  | ●        | <ul style="list-style-type: none"> <li>■ Category 0: Non-inhibitor; Category 1: inhibitor.</li> <li>■ The output value is the probability of being inhibitor, within the range of 0 to 1.</li> </ul> |

## 5. Metabolism

| Property         | Value | Decision | Comment                                                                                                                                                                  |
|------------------|-------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CYP1A2 inhibitor | 0.003 | ●        | <ul style="list-style-type: none"> <li>■ Category 1: Inhibitor; Category 0: Non-inhibitor;</li> <li>■ The output value is the probability of being inhibitor.</li> </ul> |

|                   |       |   |                                                                                                                                                                                                                                                                                                          |
|-------------------|-------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CYP1A2 substrate  | 0.0   | ● | ■ Category 1: Substrate; Category 0: Non-substrate;<br>■ The output value is the probability of being substrate.                                                                                                                                                                                         |
| CYP2C19 inhibitor | 0.007 | ● | ■ Category 1: Inhibitor; Category 0: Non-inhibitor;<br>■ The output value is the probability of being inhibitor.                                                                                                                                                                                         |
| CYP2C19 substrate | 0.001 | ● | ■ Category 1: Substrate; Category 0: Non-substrate;<br>■ The output value is the probability of being substrate.                                                                                                                                                                                         |
| CYP2C9 inhibitor  | 0.318 | ● | ■ Category 1: Inhibitor; Category 0: Non-inhibitor;<br>■ The output value is the probability of being inhibitor.                                                                                                                                                                                         |
| CYP2C9 substrate  | 0.065 | ● | ■ Category 1: Substrate; Category 0: Non-substrate;<br>■ The output value is the probability of being substrate.                                                                                                                                                                                         |
| CYP2D6 inhibitor  | 0.0   | ● | ■ Category 1: Inhibitor; Category 0: Non-inhibitor;<br>■ The output value is the probability of being inhibitor.                                                                                                                                                                                         |
| CYP2D6 substrate  | 0.0   | ● | ■ Category 1: Substrate; Category 0: Non-substrate;<br>■ The output value is the probability of being substrate.                                                                                                                                                                                         |
| CYP3A4 inhibitor  | 0.122 | ● | ■ Category 1: Inhibitor; Category 0: Non-inhibitor;<br>■ The output value is the probability of being inhibitor.                                                                                                                                                                                         |
| CYP3A4 substrate  | 0.0   | ● | ■ Category 1: Substrate; Category 0: Non-substrate;<br>■ The output value is the probability of being substrate.                                                                                                                                                                                         |
| CYP2B6 inhibitor  | 0.302 | ● | ■ Category 1: Inhibitor; Category 0: Non-inhibitor;<br>■ The output value is the probability of being inhibitor.                                                                                                                                                                                         |
| CYP2B6 substrate  | 0.0   | ● | ■ Category 1: Substrate; Category 0: Non-substrate;<br>■ The output value is the probability of being substrate.                                                                                                                                                                                         |
| CYP2C8 inhibitor  | 1.0   | ● | ■ Category 1: Inhibitor; Category 0: Non-inhibitor;<br>■ The output value is the probability of being inhibitor.                                                                                                                                                                                         |
| HLM Stability     | 0.856 | ● | ■ human liver microsomal (HLM) stability<br>■ Category 0: stable+ (HLM > 30 min); Category 1: unstable- (HLM ≤ 30 min). The output value is the probability of human liver microsomal instability, where a value closer to 1 indicates a higher likelihood of instability. The range is between 0 and 1. |

## 6. Excretion

| Property | Value | Decision | Comment |
|----------|-------|----------|---------|
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|                      |       |   |                                                                                                                                                                                                                                                                                                      |
|----------------------|-------|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CL <sub>plasma</sub> | 2.574 | ● | <p>■ The unit of predicted CL<sub>plasma</sub> penetration is ml/min/kg. &gt;15 ml/min/kg: high clearance; 5-15 ml/min/kg: moderate clearance; &lt; 5 ml/min/kg: low clearance.</p>                                                                                                                  |
| T <sub>1/2</sub>     | 2.179 | ● | <p>■ The unit of predicted T<sub>1/2</sub> is hours.</p> <p>■ ultra-short half-life drugs: 1/2 &lt; 1 hour; short half-life drugs: T<sub>1/2</sub> between 1-4 hours; intermediate short half-life drugs: T<sub>1/2</sub> between 4-8 hours; long half-life drugs: T<sub>1/2</sub> &gt; 8 hours.</p> |

## 7. Toxicity

| Property                | Value | Decision | Comment                                                                                                                                                                                                                                                                                                                                 |
|-------------------------|-------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hERG Blockers           | 0.013 | ●        | <p>■ Molecules with IC<sub>50</sub> ≤10μM or ≥50% inhibition at 10 μM were classified as hERG+ (Category 1),</p> <p>■ while molecules with IC<sub>50</sub> &gt;10μM or &lt; 50% inhibition at 10μM were classified as hERG - (Category 0).</p> <p>■ The output value is the probability of being hERG+, within the range of 0 to 1.</p> |
| hERG Blockers (10um)    | 0.141 | ●        | <p>■ Molecules with IC<sub>50</sub> ≤10 μM are classified as hERG+ (Category 1),</p> <p>■ and molecules with IC<sub>50</sub> &gt; 10μM are classified as hERG- (Category 0).</p> <p>■ The output value is the probability of being hERG+, within the range of 0 to 1.</p>                                                               |
| DILI                    | 0.999 | ●        | <p>■ Drug Induced Liver Injury.</p> <p>■ Category 1: drugs with a high risk of DILI;</p> <p>■ Category 0: drugs with no risk of DILI.</p> <p>■ The output value is the probability of being toxic.</p>                                                                                                                                  |
| AMES Mutagenicity       | 0.98  | ●        | <p>■ AMES Toxicity</p> <p>■ Category 1: Ames positive(+);</p> <p>■ Category 0: Ames negative(-);</p> <p>■ The output value is the probability of being toxic.</p>                                                                                                                                                                       |
| Rat Oral Acute Toxicity | 0.121 | ●        | <p>■ Rat Oral Acute Toxicity.</p> <p>■ Category 0: low-toxicity, &gt; 500 mg/kg;</p> <p>■ Category 1: high-toxicity; &lt; 500 mg/kg.</p> <p>■ The output value is the probability of being toxic, within the range of 0 to 1.</p>                                                                                                       |
| FDAMDD                  | 0.029 | ●        | <p>■ FDA Maximum (Recommended) Daily Dose.</p> <p>■ Category 1: FDAMDD (+);</p> <p>■ Category 0: FDAMDD (-);</p> <p>The output value is the probability of being positive.</p>                                                                                                                                                          |
| Skin Sensitization      | 1.0   | ●        | <p>■ Category 1: Sensitizer;</p> <p>■ Category 0: Non-sensitizer.</p> <p>■ The output value is the probability of being toxic, within the range of 0 to 1.</p>                                                                                                                                                                          |
| Carcinogenicity         | 0.5   | ●        | <p>■ Category 1: carcinogens;</p> <p>■ Category 0: non-carcinogens;</p> <p>■ The output value is the probability of being toxic.</p>                                                                                                                                                                                                    |

|                               |       |                                                                                     |                                                                                                                                                                                             |
|-------------------------------|-------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Eye Corrosion                 | 0.006 |    | <p>■ Eye Corrosion</p> <p>■ Category 1: corrosives;</p> <p>Category 0: noncorrosives;</p> <p>■ The output value is the probability of being corrosives.</p>                                 |
| Eye Irritation                | 0.854 |    | <p>■ Eye Irritation</p> <p>■ Category 1: irritants;</p> <p>Category 0: nonirritants;</p> <p>■ The output value is the probability of being irritants.</p>                                   |
| Respiratory                   | 0.053 |    | <p>■ Category 1: respiratory toxicants;</p> <p>■ Category 0: non-respiratory toxicants.</p> <p>■ The output value is the probability of being toxic, within the range of 0 to 1.</p>        |
| Human Hep atotoxicity         | 0.638 |    | <p>■ Human Hepatotoxicity</p> <p>■ Category 1: H-HT positive(+);</p> <p>■ Category 0: H-HT negative(-);</p> <p>■ The output value is the probability of being toxic.</p>                    |
| Drug-induce d Nephrotox icity | 0.801 |    | <p>■ Category 0: non-nephrotoxic (-);</p> <p>■ Category 1: nephrotoxic (+).</p> <p>■ The output value is the probability of being nephrotoxic (+), within the range of 0 to 1.</p>          |
| Ototoxicity                   | 0.684 |  | <p>■ Category 0: non-ototoxicity (-);</p> <p>■ Category 1: ototoxicity (+).</p> <p>■ The output value is the probability of being ototoxicity (+), within the range of 0 to 1.</p>          |
| Hematotoxic ity               | 0.647 |  | <p>■ Category 0: non-hematotoxicity (-);</p> <p>■ Category 1: hematotoxicity (+).</p> <p>■ The output value is the probability of being hematotoxicity (+), within the range of 0 to 1.</p> |
| Genotoxicity                  | 1.0   |  | <p>■ Category 0: non-Genotoxicity (-);</p> <p>■ Category 1: Genotoxicity (+).</p> <p>■ The output value is the probability of being ototoxicity (+), within the range of 0 to 1.</p>        |
| RPMI-8226 Immunitoxici ty     | 0.123 |  | <p>■ Category 0: non-cytotoxicity (-);</p> <p>■ Category 1: cytotoxicity (+).</p> <p>■ The output value is the probability of being ototoxicity (+), within the range of 0 to 1.</p>        |
| A549 Cytotoxicity             | 0.742 |  | <p>■ Category 0: non-cytotoxicity (-);</p> <p>■ Category 1: cytotoxicity (+).</p> <p>■ The output value is the probability of being ototoxicity (+), within the range of 0 to 1.</p>        |
| Hek293 Cytotoxicity           | 0.227 |  | <p>■ Category 0: non-cytotoxicity (-);</p> <p>■ Category 1: cytotoxicity (+).</p> <p>■ The output value is the probability of being ototoxicity (+), within the range of 0 to 1.</p>        |
| Drug-induce d Neurotox icity  | 0.006 |  | <p>■ Category 0: non-neurotoxic (-);</p> <p>■ Category 1: neurotoxic (+).</p> <p>■ The output value is the probability of being neurotoxic (+), within the range of 0 to 1.</p>             |

## 8. Environmental toxicity

| Property                 | Value | Comment                                                                                                                                                                                                                                                                          |
|--------------------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bioconcentration Factors | 0.469 | <ul style="list-style-type: none"> <li>■ Bioconcentration factors are used for considering secondary poisoning potential and assessing risks to human health via the food chain.</li> <li>■ The unit is <math>-\log_{10}[(\text{mg/L})/(1000 \cdot \text{MW})]</math></li> </ul> |
| IGC <sub>50</sub>        | 3.236 | <ul style="list-style-type: none"> <li>■ Tetrahymena pyriformis 50 percent growth inhibition concentration.</li> <li>■ The unit is <math>-\log_{10}[(\text{mg/L})/(1000 \cdot \text{MW})]</math></li> </ul>                                                                      |
| LC <sub>50</sub> FM      | 4.057 | <ul style="list-style-type: none"> <li>■ 96-hour fathead minnow 50 percent lethal concentration.</li> <li>■ The unit is <math>-\log_{10}[(\text{mg/L})/(1000 \cdot \text{MW})]</math></li> </ul>                                                                                 |
| LC <sub>50</sub> DM      | 4.49  | <ul style="list-style-type: none"> <li>■ 48-hour daphnia magna 50 percent lethal concentration.</li> <li>■ The unit is <math>-\log_{10}[(\text{mg/L})/(1000 \cdot \text{MW})]</math></li> </ul>                                                                                  |

## 9. Tox21 pathway

| Property      | Value | Decision | Comment                                                                                                                                                                                                                                 |
|---------------|-------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NR-AhR        | 0.097 | ●        | <ul style="list-style-type: none"> <li>■ Aryl hydrocarbon receptor</li> <li>■ Category 1: actives ;</li> <li>■ Category 0: inactives;</li> <li>■ The output value is the probability of being active.</li> </ul>                        |
| NR-AR         | 0.027 | ●        | <ul style="list-style-type: none"> <li>■ Androgen receptor</li> <li>■ Category 1: actives ;</li> <li>■ Category 0: inactives;</li> <li>■ The output value is the probability of being active.</li> </ul>                                |
| NR-AR-LBD     | 0.021 | ●        | <ul style="list-style-type: none"> <li>■ Androgen receptor ligand-binding domain</li> <li>■ Category 1: actives ;</li> <li>■ Category 0: inactives;</li> <li>■ The output value is the probability of being active.</li> </ul>          |
| NR-Aromatase  | 0.016 | ●        | <ul style="list-style-type: none"> <li>■ Category 1: actives ;</li> <li>■ Category 0: inactives;</li> <li>■ The output value is the probability of being active.</li> </ul>                                                             |
| NR-ER         | 0.25  | ●        | <ul style="list-style-type: none"> <li>■ Estrogen receptor</li> <li>■ Category 1: actives ;</li> <li>■ Category 0: inactives;</li> <li>■ The output value is the probability of being active.</li> </ul>                                |
| NR-ER-LBD     | 0.005 | ●        | <ul style="list-style-type: none"> <li>■ Estrogen receptor ligand-binding domain</li> <li>■ Category 1: actives ;</li> <li>■ Category 0: inactives;</li> <li>■ The output value is the probability of being active.</li> </ul>          |
| NR-PPAR-gamma | 0.0   | ●        | <ul style="list-style-type: none"> <li>■ Peroxisome proliferator-activated receptor gamma</li> <li>■ Category 1: actives ;</li> <li>■ Category 0: inactives;</li> <li>■ The output value is the probability of being active.</li> </ul> |
| SR-ARE        | 0.208 | ●        | <ul style="list-style-type: none"> <li>■ Antioxidant response element</li> <li>■ Category 1: actives ;</li> <li>■ Category 0: inactives;</li> <li>■ The output value is the probability of being active.</li> </ul>                     |



|          |       |   |                                                                                                                                                                                                                                      |
|----------|-------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SR-ATAD5 | 0.001 | ● | <ul style="list-style-type: none"> <li>■ ATPase family AAA domain-containing protein 5</li> <li>■ Category 1: actives ;</li> <li>■ Category 0: inactives;</li> <li>■ The output value is the probability of being active.</li> </ul> |
| SR-HSE   | 0.04  | ● | <ul style="list-style-type: none"> <li>■ Heat shock factor response element</li> <li>■ Category 1: actives ;</li> <li>■ Category 0: inactives;</li> <li>■ The output value is the probability of being active.</li> </ul>            |
| SR-MMP   | 0.179 | ● | <ul style="list-style-type: none"> <li>■ Mitochondrial membrane potential</li> <li>■ Category 1: actives ;</li> <li>■ Category 0: inactives;</li> <li>■ The output value is the probability of being active.</li> </ul>              |
| SR-p53   | 0.112 | ● | <ul style="list-style-type: none"> <li>■ p53, a tumor suppressor protein</li> <li>■ Category 1: actives ;</li> <li>■ Category 0: inactives;</li> <li>■ The output value is the probability of being active.</li> </ul>               |

## 10. Toxicophore Rules

| Property                          | Value    | Comment                                                                                                                          |
|-----------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------|
| Acute Toxicity Rule               | 0        | <ul style="list-style-type: none"> <li>■ 20 substructures;</li> <li>■ acute toxicity during oral administration</li> </ul>       |
| Genotoxic Carcinogenicity Rule    | 5 alerts | <ul style="list-style-type: none"> <li>■ 117 substructures;</li> <li>■ carcinogenicity or mutagenicity</li> </ul>                |
| NonGenotoxic Carcinogenicity Rule | 0        | <ul style="list-style-type: none"> <li>■ 23 substructures;</li> <li>■ carcinogenicity through nongenotoxic mechanisms</li> </ul> |
| Skin Sensitization Rule           | 5 alerts | <ul style="list-style-type: none"> <li>■ 155 substructures;</li> <li>■ skin irritation</li> </ul>                                |
| Aquatic Toxicity Rule             | 0        | <ul style="list-style-type: none"> <li>■ 99 substructures;</li> <li>■ toxicity to liquid(water)</li> </ul>                       |
| NonBiodegradable Rule             | 3 alerts | <ul style="list-style-type: none"> <li>■ 19 substructures;</li> <li>■ non-biodegradable</li> </ul>                               |
| SureChEMBL Rule                   | 0        | <ul style="list-style-type: none"> <li>■ 164 substructures;</li> <li>■ MedChem unfriendly status</li> </ul>                      |
| FAF-Drugs4 Rule                   | 3 alerts | 154 toxic substructures from FAF-Drug4                                                                                           |